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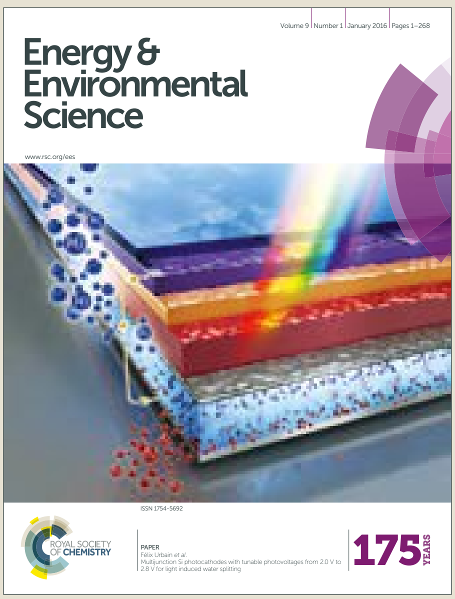

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COMMUNICATION

2D boron nitride nanoflakes as a multifunctional additive in gel polymer electrolyte for safe, long cycle life and high rate lithium metal batteries

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The multifunctional properties of 2D boron nitride nanoflakes (BNNFs) in gel polymer electrolyte (GPE) are presented. A small addition (0.5 wt%) of BNNFs into GPE substantially improves all the advantageous properties of the GPE including ionic conductivity, Li^+ transference number, mechanical modulus, and dendrite-suppressing capability, thereby enabling high performance lithium metal batteries.

Broader context

With the growing demand for energy storage technologies meeting the requirements of emerging electric vehicles and grid-scale energy storage systems, researches on rechargeable batteries with enhanced safety, longer cycle life, and higher energy and power densities are becoming increasingly important. Lithium metal batteries (LMBs) are thus of great interest as one of the most promising technologies for energy storage applications, owing to the high theoretical specific capacity (3860 mAh g^{-1}), low density (0.59 g cm^{-3}), and low redox potential (–3.04 V *vs.* the standard hydrogen electrode) of the lithium (Li) metal anode. However, poor cycle life and serious safety hazards of the LMBs originating from the uncontrollable formation and growth of Li dendrites during charge/discharge cycles have hampered their practical applications. Extensive efforts have been made to suppress the Li dendrites, while several limitations in terms of constrained performance and/or complicated processes still need to be addressed. Herein, an effective GPE for safe, long cycle life and high rate LMBs is simply prepared using BNNFs as a multifunctional additive. This study offers a fundamental insight into the application of multifunctional BNNFs and a facile approach for preparing dendrite-suppressing GPEs with potential applicability in next-generation LMBs, including Li/S and Li/Air batteries.

Lithium metal batteries (LMBs), which utilize metallic lithium (Li) as the anode, have received considerable attention as an attractive alternative to conventional lithium-ion batteries (LIBs) for the applications in electric vehicles and energy storage systems.[1] Since Li metal, the lightest metal among all the metallic elements, possesses the highest theoretical specific capacity (3860 mAh g^{-1}) and the lowest redox potential (–3.04 V *vs.* the standard hydrogen electrode), high energy density and high voltage batteries can be prepared using the Li metal anode system.[1b] Moreover, Li metal anode enables the use of high capacity, unlithiated cathode materials such as sulfur and oxygen as well as the elimination of the current collectors used in conventional anodes.[1,2] However, practical applications of LMBs are currently limited by the formation and growth of Li dendrites in an uncontrollable way during repeated charge/discharge cycles, which lead to short cycle life and serious safety issues including internal short circuits and thermal runaway of the batteries.[2]

To control the Li dendrites, several strategies including *in situ* formation of stable solid-electrolyte interphase (SEI) layers[3] or self-healing electrostatic shield[4] by electrolyte additives, *ex situ* protective coating of the Li anode[5], and application of solid electrolytes with high modulus and high Li^+ transference number (t_{Li+})[6] have been investigated over the years. In general, there are two main theoretical frameworks for understanding the suppression of Li dendrites. One model by Chazalviel *et. al.* suggests that higher ionic conductivity and higher t_{Li+} of electrolytes can restrain the nucleation of Li dendrites by mitigating anion depletion-induced large electric fields near the Li electrode.[7] The other is the Monroe and Newman model that describes the suppression of Li dendrite growth by mechanical blocking using electrolytes with high shear modulus about twice that of the Li metal.[8] Based on these theoretical frameworks, the use of solid electrolytes with high modulus and high t_{Li+} is considered as one of the most promising approaches. However, the solid electrolytes have intrinsic drawbacks for practical applications, such as insufficient ionic conductivities at room temperature for solid polymer electrolytes and difficult fabrication and handling processes for solid ceramic electrolytes.[9] Although gel polymer electrolytes (GPEs) are suitable to resolve these issues by taking advantages of their ease of fabrication, high ionic conductivity, and excellent electrochemical performance,

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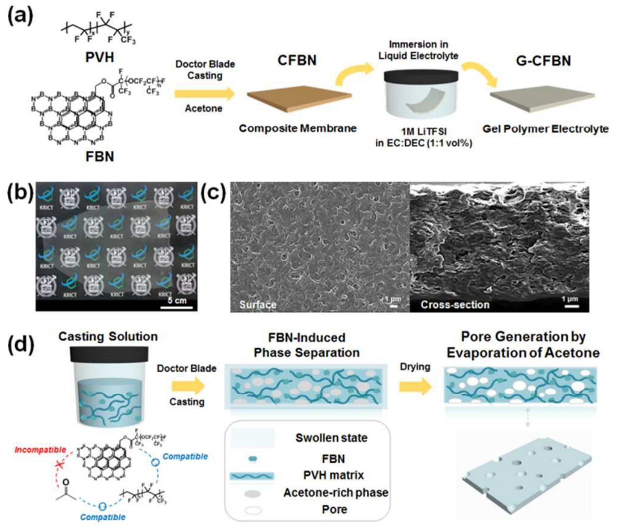

Fig. 1 Preparation of PVH-based GPE with FBN additive (G-CFBN). (a) Schematic illustration of the overall procedure for preparation of G-CFBN, (b) photograph of CFBN (0.5 wt% FBN), (c) surface (left) and cross-sectional (right) SEM images of CFBN (0.5 wt% FBN), and (d) suggested mechanism on autonomous pore formation of CFBN by FBN-induced phase separation.

there have been only a few reports concerning Li dendrite suppression by GPEs.[10] In order to utilize the GPEs for LMB applications, their ionic conductivity and mechanical modulus have been further improved by introducing inorganic fillers, such as SiO_2, Al_2O_3, and TiO_2.[10e,11] However, in most cases, a large amount of fillers were required to mechanically block the Li dendrite growth, while also sacrificing the ionic conductivity.[10b] Most fillers are easily aggregated when the filler content exceeds a certain value, and then the aggregated filler domains deteriorate well-defined ion-conducting path, resulting in decreased ionic conductivity.[10b] To the best of our knowledge, there has been no report demonstrating the fabrication of Li dendrite-suppressing GPEs with most of the desirable properties such as high ionic conductivity, high t_{Li+}, and high shear modulus *via* small addition (< 1 wt%) of inorganic fillers.

Recently, 2D nanomaterials, such as graphene and graphene analogues, have been studied extensively for a wide range of applications, due to their attractive properties originating from the ultrathin structure with a high degree of anisotropy and chemical functionality.[12] Especially, the large enhancement of physical and chemical properties of diverse composites has been reported even with slight addition of the 2D nanomaterials as functional nanofillers, which is ascribed to their exceptionally large interactive surface area providing efficient interfacial interactions with the polymer matrices.[13] Few layered hexagonal boron nitride (BN), as one of the graphene analogues, not only shares common properties with it, such as high mechanical strength and thermal conductivity, but also exhibits other interesting properties owing to its unique structural features.[14] In contrast to the C-C bonds of graphene, the B-N bonds of BN show partially ionic character; the higher electronegativity of N atoms makes electron pairs in sp^2-hybridized B-N σ bonds more confined to the N atoms.[12a,14] Also, the lone pair electrons in the N p_z orbital are only partially delocalized with the empty B p_z orbital.[12a,14] These characteristics lead to the intrinsic electrically insulating property and superior thermal and electrochemical stability of the BN,[12a,14] which are prerequisites for the electrolyte applications. In addition, B atoms in BN with Lewis acidic characteristics can interact with other Lewis bases,[15] thereby expected to increase the t_{Li+} of GPEs by trapping the anions of electrolytes. This feature also makes BN an attractive candidate for GPE additives, while it has not yet been fully elucidated in the literature.

There have been a few reports on the battery applications of the BN, presenting its use in ionic liquid-based composite electrolytes[16] and stable coating layers on Li metal anode[17] and separator[18], but its remarkable ability for suppressing the Li dendrites as a GPE additive has not been explored yet. Here, we report for the first time a simple and effective strategy to prepare a Li dendrite-suppressing GPE using perfluoropolyether (PFPE)-functionalized 2D BN nanoflakes (BNNFs) as a multifunctional additive. It is demonstrated that even a minimal addition (0.5 wt%) of the PFPE-functionalized BNNFs into the GPE can provide an unprecedented combination of high ionic conductivity, high t_{Li+}, and high mechanical modulus, which all contribute to the effective suppression of Li dendrites for LMBs with excellent electrochemical and safety performance.

In order to increase the surface area and compatibility with P(VdF-*co*-HFP) (PVH) GPE matrix, PFPE-functionalized BNNFs (FBN) were prepared *via* sonication-assisted exfoliation and noncovalent functionalization of nano-sized BN powder using PFPE-functionalized pyrene molecules (Fig. 1a and Fig. S1-S5, ESI†). The FBN can maintain the 2D morphology with a much decreased thickness of 3–4 nm, corresponding to 9–12 BN layers, even after the exfoliation and PFPE functionalization (≈4.7 wt%) (Fig. S6-S9, ESI†). Composite membranes (CFBNs) were fabricated *via* simple blade casting of the mixture of PVH and FBN in acetone on a glass plate, wherein the FBN content was controlled as 0.1, 0.2, 0.5, and 1.0 wt% of the PVH. After dried at ambient condition and subsequently under vacuum at 60 ^{o}C, porous CFBNs with thickness of 6–9 μm could be obtained, which were further immersed in liquid electrolyte (1M LiTFSI in EC:DEC (1:1 vol%)) to prepare GPEs (G-CFBNs) (Fig. 1a-c). It is worth noting that the porous structure of CFBNs is automatically generated during the membrane fabrication process without the addition of any porogen or non-solvent. This should be ascribed to the different miscibility behaviour of FBN with the PVH matrix and the acetone solvent (Fig. 1d). PVH is fully soluble in acetone, while the FBN with hydrophobic PFPE chains is not soluble and even cannot be dispersed in acetone. However, FBN was found to maintain its dispersity in acetone for 1 h when mixed with PVH (Fig. S11, ESI†). The miscibility between FBN and PVH is expected, because PVH and the PFPE chains in FBN are all fluorinated polymers.[19] Thus, the PFPE chains of FBN could provide selective compatibility with the PVH matrix rather than the

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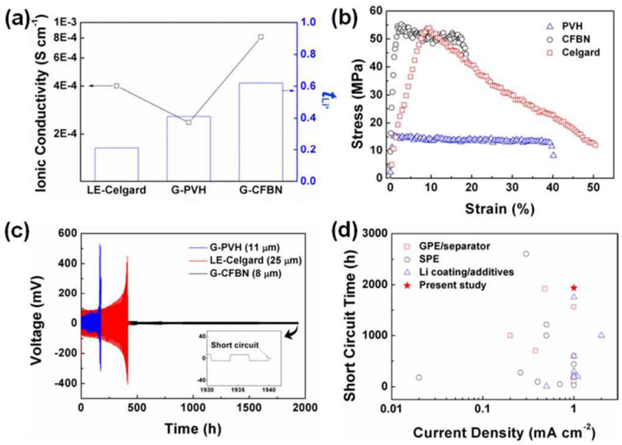

Fig. 2 Effect of FBN on electrochemical and mechanical properties. (a) Ionic conductivity and Li^+ transference number (t_{Li+}) of LE-Celgard, G-PVH, and G-CFBN at 25 °C, (b) stress-strain curves of PVH, CFBN, and Celgard, (c) galvanostatic cycling profiles of symmetric Li/Li cells with LE-Celgard, G-PVH, and G-CFBN at a current density of 1.0 mA cm^{-2} at 25 °C, and (d) short circuit time of G-CFBN compared with other state of the art dendrite-suppressing strategies at various current densities.

acetone, which induces autonomous phase separation between the FBN-incorporated PVH matrix and the acetone during the membrane casting process. Porous membrane structure is then generated by evaporation of the phase-separated solvent. In contrast to the non-porous, dense membrane structure obtained from the PVH without FBN, the CFBNs exhibit porous structures where the pore size gradually increases with the FBN content, possibly due to the increase of aforementioned phase separation (Fig. S12, ESI†).

We investigated the advantageous effects of FBN in GPE by analysing the electrochemical and mechanical properties of G-CFBN, PVH-based GPE without FBN (G-PVH), and liquid electrolyte system containing a commercialized separator (LE-Celgard) (Fig. S10, ESI†). The G-CFBN containing 0.5 wt% FBN of the PVH matrix was used for the analysis, because it exhibits the highest ionic conductivity among the G-CFBNs prepared in this study (Fig. S15, ESI†). The ionic conductivity of G-CFBN increases as the FBN content increases from 0 wt% (G-PVH) to 0.5 wt%, due to the increase of electrolyte uptake. The incorporation of FBN into PVH matrix increases the electrolyte uptake, because it decreases the crystallinity of the PVH and generates the porous structure (Fig. S12 and S13, ESI†).[10b] When the FBN content is larger than 0.5 wt%, the electrolyte uptake in G-CFBN was found to decrease, which in turn decreases the ionic conductivity. This could be attributed to the formation of crystalline structure in the PVH matrix that can decrease the effective electrolyte uptake (Fig. S13, ESI†),[10b,20] which originates from the inhomogeneous dispersion and the aggregation of excess FBN (Fig. S12e, ESI†).[20] It should be noted that the ionic conductivity of G-CFBN containing only 0.5 wt% of FBN is about twice higher than that of the conventional LE-Celgard (Fig. 2a). Meanwhile, a GPE (G-CBN) with 0.5 wt% pristine BN that is incompatible with PVH shows crystalline phases and non-porous, dense morphology in the PVH matrix (Fig. S10c and S14, ESI†). This leads to lower electrolyte uptake and ionic conductivity as compared to the G-CFBN (Fig. S16, ESI†).

The anion-trapping ability of FBN was evaluated by measuring the t_{Li+} of G-CFBN, G-PVH, and LE-Celgard, using DC polarization/AC impedance combination method (Fig. 2a and Fig. S17, ESI†). As the fluorinated polymer chains in G-PVH can interact with anions in the electrolyte due to their high dielectric constant (9.0-10.0) and electron-withdrawing functional groups, the G-PVH without the FBN already exhibits larger t_{Li+} value than the LE-Celgard.[21] Although the anion-trapping ability of BN in electrolytes has not yet been reported, the surface functionalization of BN by the interaction between electron-deficient B atoms in BN and Lewis bases has been reported previously.[15b,15c] The higher t_{Li+} value of G-CFBN (0.62) than that of the G-PVH (0.41) clearly demonstrates that the FBN in G-CFBN can work as a Lewis acid to trap the anions of the electrolyte, although the empty B p_z orbital is partially filled with delocalized lone pair electrons in the N p_z orbital.[12a,15b] The anion-trapping ability of FBN in G-CFBN could be further confirmed by Raman spectra of the GPEs (Fig. S18, ESI†). The intensity of Li^+-uncoordinated $TFSI^-$ peak at 742 cm^{-1} in G-CFBN was found to be larger than that in G-PVH, indicating that the FBN interacts with the $TFSI^-$ and thus dissociates it from the Li^+-coordinated state.[22] Such interaction between the FBN and $TFSI^-$ also contributes to the larger electrochemical stability window of G-CFBN than the G-PVH by lowering the extent of anion oxidation at high potential (Fig. S19, ESI†).[22,23] Therefore, with only small content of FBN such as 0.5 wt%, the G-CFBN exhibits much improved overall electrochemical properties including the ionic conductivity and t_{Li+}, which are important in controlling the Li dendrite formation by alleviating the anion depletion-induced large electric fields near the Li anode.[7]

2D nanomaterials with large surface areas are known to have the effective reinforcement ability in the composites *via* efficient interfacial interactions with polymer matrices.[13,14,24] The only small amount of FBN incorporated into the PVH matrix also significantly enhances the mechanical strength (Fig. 2b). Young's modulus (110 GPa) and tensile strength (53 MPa) of CFBN, analyzed by universal testing machine (UTM), are 4.4 and 3.2 times larger than those of PVH, respectively (Fig. S22, ESI†). In addition, the Young's modulus of the CFBN is 70 times higher than that of the Celgard, although their tensile strengths are close. Theoretically, high shear modulus (G) of GPEs, larger than 7 GPa, is considered as an important physical parameter for impeding the Li dendrite growth.[8] The G of the CFBN can be estimated from the equation, $G = E/[2(1 + v)]$, where E and v are the Young's modulus and Poisson's ratio, respectively.[25] Since the v typically ranges from –1 to 0.5,[25] the CFBN should possess G higher than at least 36.7 GPa. The E values of GPEs after the equilibrated uptake of liquid electrolyte could not be measured by the UTM, because the samples for the measurement were excessively slippery. However, it might be informative that E value of 27.4 GPa could be obtained for the G-CFBN from repeated

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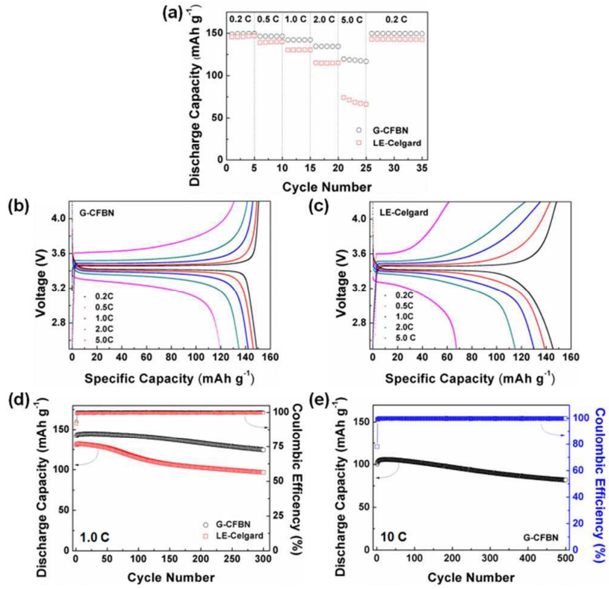

Fig. 3 Electrochemical performance of Li/electrolyte/$LiFePO_4$ cells cycled at 25 °C, where the electrolyte is G-CFBN and LE-Celgard. (a) Rate capability of the cells at various C-rates and corresponding voltage-capacity curves of the cells containing (b) G-CFBN and (c) LE-Celgard, (d) long-term cycling performance of the cells at 1.0 C, and (e) long-term cycling performance of the cell containing G-CFBN at 10 C.

measurements, when the electrolyte uptake was intentionally limited to about 60% by controlling the uptake time (Fig. S23, ESI†). The 60% of electrolyte uptake was the maximum amount for measuring the mechanical property without any slippery behaviour. Thus, we believe that G-CFBN has a large enough mechanical modulus value for suppressing the Li dendrite growth as a GPE in LMBs.

The formation and growth of Li dendrites in symmetric Li/Li cells with G-PVH, G-CFBN, and LE-Celgard were compared by performing galvanostatic Li plating/stripping cycling tests at a high current density of 1 mA cm^{-2} (Fig. 2c). The Li/G-PVH(11 μm thickness)/Li cell shows a short circuit induced by Li dendrite growth, which is characterized by the sharp rise in voltage hysteresis with the cycle time and the sudden drop after 178 h.[10b] Since the smaller short circuit time of Li/G-PVH(11 μm thickness)/Li cell than that of Li/LE-Celgard(25 μm thickness)/Li cell might be caused by the thinner G-PVH thickness, Li/G-PVH(24 μm thickness)/Li cell was also tested; still the G-PVH based cell shows smaller short circuit time than the LE-Celgard based cell (Fig. S20, ESI†). The non-porous structure and crystallinity of the PVH could cause limited, inhomogeneous electrolyte uptake, resulting in the facilitated Li dendrite growth by non-uniform Li^+ flux near the Li anode.[26] However, the Li/Li cell with thinner G-CFBN (8 μm) exhibits much smaller overpotential in the voltage hysteresis and longer cycling stability (1940 h) without any short circuit than those with G-PVH and LE-Celgard, which should be attributed to all the desirable properties of high ionic conductivity, high t_{Li+}, and large mechanical modulus of the G-CFBN. The excellent compatibility between the G-CFBN and Li electrode is also supported by the smooth Li electrode surface morphology after the short circuit test (Fig. S21, ESI†).[10b,10c] Despite extensive efforts *via* various strategies, the Li dendrite suppression, especially at high current densities (>1 mA cm^{-2}), still remains a challenge.[27] As compared with representative results across recently published studies, the G-CFBN clearly exhibits comparable or somewhat superior Li dendrite-suppressing ability at a high current density of 1 mA cm^{-2} (Fig. 2d and Table S1, ESI†, for details). Therefore, it is shown here that a promising GPE for dendrite-free LMBs can be prepared *via* a simple and effective approach of utilizing 2D BNNFs as a multifunctional additive.

We further evaluated the practical applicability of G-CFBN in LMBs using a Li metal anode and a $LiFePO_4$ (LFP) cathode. The Li/G-CFBN/LFP cell shows superior rate capability in comparison with the Li/LE-Celgard/LFP cell especially at a high C-rate of 5.0 C (Fig. 3a-3c, ESI†), mainly due to the higher ionic conductivity and t_{Li+} of the G-CFBN than those of the LE-Celgard (Fig. 2a).[22] As shown in the long-term cycling performance at 1.0 C (Fig. 3d and Fig. S26, ESI†), the cell with G-CFBN also exhibits higher capacity retention (88 %) than that with LE-Celgard (74 %) after 300 cycles. In addition, overpotential values of the Li/G-CFBN/LFP cell are much smaller than those of Li/LE-Celgard/LFP cell in voltage–capacity profiles (Fig. S26, ESI†), indicating that problematic electrode polarization is significantly suppressed by the G-CFBN. Upon disassembling the cells after the cycling, Li metal surface in the cell with G-CFBN shows somewhat porous but smooth morphology, whereas that in the cell with LE-Celgard shows noticeably large cracks and pinholes with roughness (Fig. S27, ESI†), demonstrating that the dendrite growth is much suppressed by the G-CFBN compared to the LE-Celgard. This should be attributed to the synergistic combination of superior mechanical modulus[25] and excellent electrochemical properties[10c,27] of the G-CFBN (Fig. 2). The controlled Li plating/stripping behaviour eventually leads to the formation of stable SEI layer and thereby achieves better long-term cycling performance.[27] The more stable SEI formation for the cell with the G-CFBN is further verified from the electrochemical impedance spectroscopy (EIS) of each cell before and after cycling (Fig. S28, ESI†), presenting smaller interfacial resistance (R_{int} = R_{SEI} + R_{ct}) values of the Li/G-CFBN/LFP cell as compared to the Li/LE-Celgard/LFP cell. Moreover, it is noteworthy that the Li/G-CFBN/LFP cell exhibits outstanding cycling performance over 500 cycles even when the C-rate reaches 10 C (Fig. 3e and Fig. S29, ESI†); to our knowledge, this can be considered as an unprecedented high-rate long-term cycling performance compared with other reports in the literature (Table S2, ESI†). In contrast, the Li/LE-Celgard/LFP cell exhibits very poor cycling performance at 10 C (Fig. S30, ESI†). The outstanding long-term cycling stability at high C-rates, combined with intrinsic good thermal stability of the CFBN matrix (Fig. S24, ESI†), can eventually lead to realization of safe, long cycle life and high rate LMBs with the G-CFBN.

In summary, we have demonstrated that the PFPE-functionalized 2D BNNFs (FBN) additive can provide

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multifunctional properties to a PVH-based GPE for LMB applications. Even with the very small content of FBN such as 0.5 wt%, the GPE containing FBN (G-CFBN) exhibits greatly enhanced overall electrochemical and physical properties, including ionic conductivity, t_{Li+}, and mechanical modulus. Such high ionic conductivity, high t_{Li+}, and high mechanical modulus make the G-CFBN strongly resistant against the Li dendrite formation and growth by providing both alleviated Li^+ concentration gradient and mechanically robust blocking layer. The G-CFBN in symmetric Li/Li cells shows unprecedentedly long short circuit time of 1940 h at a high current density of 1 mA cm^{-2}. The G-CFBN also works efficiently in Li/LFP batteries, where they can be reversibly cycled maintaining high discharge capacity for more than 500 cycles, even at a high 10 C rate. We believe that the present work provides insight into the design and preparation of dendrite-suppressing GPEs for LMB applications including next-generation Li/S and Li/Air batteries, by taking advantages of the multifunctional properties of 2D BNNFs. In particular, the Lewis acidity of BNNFs in GPEs could exhibit additional polysulfide trapping ability for controlling the polysulfide dissolution in Li/S batteries; such studies are underway.

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Notes and references

1 (*a*) K. Zhang, G.-H. Lee, M. Park, W. Li, Y.-M. Kang, *Adv. Energy Mater.* **2016**, *6*, 1600811; (*b*) W. Xu, J. Wang, F. Ding, X. Chen, E. Nasybulin, Y. Zhang, J.-G. Zhang, *Energy Environ. Sci.* **2014**, *7*, 513-537; (*c*) H.-G. Wang, S. Yuan, D.-L. Ma, X.-B. Zhang, J.-M. Yah, *Energy Environ. Sci*. **2015**, *8*, 1660-1681; (*d*) T. Sun, Z.-J. Li, H.-G. Wang, D. Bao, F.-L. Meng, X.-B. Zhang, *Angew. Chem*. **2016**, *128*, 10820-10824.

2 D. Lin, Y. Liu, Y. Cui, *Nat. Nanotechnol.* **2017**, *12*, 194-206.

3 (*a*) Y. Lu, Z. Tu, L. A. Archer, *Nat. Mater.* **2014**, *13*, 961-969; (*b*) S. Choudhury, L. A. Archer, *Adv. Electron. Mater.* **2016**, *2*, 1500246.

4 F. Ding, W. Xu, G. L. Graff, J. Zhang, M. L. Sushko, X. Chen, Y. Shao, M. H. Engelhard, Z. Nie, J. Xiao, X. Liu, P. V. Sushko, J. Liu, J.-G. Zhang, *J. Am. Chem. Soc.* **2013**, *135*, 4450-4456.

5 (*a*) E. Kazyak, K. N. Wood, N. P. Dasgupta, *Chem. Mater.* **2015**, *27*, 6457-6462; (*b*) D. Lin, Y. Liu, Z. Liang, H.-W. Lee, J. Sun, H. Wang, K. Yan, J. Xie, Y. Cui, *Nat. Nanotechnol.* **2016**, *11*, 626-632; (*c*) D. Lin, D. Zhuo, Y. Liu, Y. Cui, *J. Am. Chem. Soc*. **2016**, *138*, 11044-11050; (*d*) Q.-C. Liu, T. Liu, D.-P. Liu, Z.-J. Li, X.-B. Zhang, Y. Zhang, *Adv. Mater*. **2016**, *28*, 8413-8418; (*e*) T. Liu, Q.-C. Liu, J.-J. Xu, X.-B. Zhang, *Small* **2016**, *12*, 3101-3105; (*f*) G. Zheng, S. W. Lee, Z. Liang. H.-W. Lee, K. Yan, H. Yao, H. Wang, W. Li, S. Chu, Y. Cui, *Nat. Nanotechnol*. **2014**, *9*, 618-623.

6 (*a*) S.-O. Tung, S. Ho, M. Yang, R. Zhang, N. A. Kotov, *Nat. Commun.* **2015**, *6*, 6152; (*b*) Q. Pan, D. M. Smith, H. Qi, S. Wang, C. Y. Li, *Adv. Mater.* **2015**, *27*, 5995-6001.

7 J.-N. Chazalviel, *Phys. Rev. A* **1990**, *42*, 7355-7367.

8 C. Monroe, J. Newman, *J. Electrochem. Soc.* **2005**, *152*, A396-A404.

9 A. Varzi, R. Raccichini, S. Passerini, B. Scrosati, *J. Mater. Chem. A* **2016**, *4*, 17251-17259.

10 (*a*) Q. Lu, Y.-B. He, Q. Yu, B. Li, Y. V. Kaneti, Y. Yao, F. Kang, Q.-H. Yang, *Adv. Mater.* **2017**, 1604460; (*b*) H. T. T. Le, D. T. Ngo, R. S. Kalubarme, G. Cao, C.-N. Park, C.-J. Park, *ACS Appl. Mater. Interfaces* **2016**, *8*, 20710-20719; (*c*) D. Zhou, R. Liu, Y.-B. He, F. Li, M. Liu, B. Li, Q.-H. Yang, Q. Cai, F. Kang, *Adv. Energy Mater.* **2016**, *6*, 1502214; (*d*) Y. Zhu, S. Xiao, Y. Shi, Y. Yang, Y. Hou, Y. Wu, *Adv. Energy Mater.* **2014**, *4*, 1300647; (*e*) S. Choudhury, R. Mangal, A. Agrawal, L. A. Archer, *Nat. Commun.* **2015**, *6*, 10101.

11 (*a*) L. Long, S. Wang, M. Xiao, Y. Meng, *J. Mater. Chem. A* **2016**, *4*, 10038-10069; (*b*) N. Byrne, J. Efthimiadis, D. R. MacFarlane, M. Forsyth, J. Mater. Chem. 2004, 14, 127-133.

12 (*a*) Q. Weng, X. Wang, X. Wang, Y. Bando, D. Golberg, *Chem. Soc. Rev.* **2016**, *45*, 3989-4012; (*b*) K. S. Novoselov, V. I. Fal'ko, L. Colombo, P. R. Gellert, M. G. Schwab, K. Kim, *Nature* **2012**, *490*, 192-200.

13 (*a*) T. Kuilla, S. Bhadra, D. Yao, N. H. Kim, S. Bose, J. H. Lee, *Prog. Polym. Sci.* **2010**, *35*, 1350-1375; (b) M.-Y. Lim, H. J. Kim, S. J. Baek, K. Y. Kim, S.-S. Lee, J.-C. Lee, *Carbon* **2014**, 77, 366-378; (c) J. Shim, D.-G. Kim, H. J. Kim, J. H. Lee, J.-H. Baik, J.-C. Lee, *J. Mater. Chem. A* **2014**, 2, 13873-13883.

14 A. Pakdel, Y. Bando, D. Golberg, *Chem. Soc. Rev.* **2014**, *43*, 934-959.

15 (*a*) K.-H. Oh, D. Lee, M.-J. Choo, K. H. Park, S. Jeon, S. H. Hong, J.-K. Park, J. W. Choi, *ACS Appl. Mater. Interfaces* **2014**, *6*, 7751-7758; (*b*) S.-Y. Xie, W. Wang, K. A. S. Fernando, X. Wang, Y. Lin, Y.-P. Sun, *Chem. Commun.* **2005**, 3670-3672; (*c*) Y. Lin, T. V. Williams, J. W. Connell, *J. Phys. Chem. Lett.* **2010**, *1*, 277-283.

16 (*a*) M.-T. F. Rodrigues, K. Kalaga, H. Gullapalli, G. Babu, A. L. M. Reddy, P. M. Ajayan, *Adv. Energy Mater.* **2016**, *6*, 1600218; (*b*) M. Li, W. Zhu, P. Zhang, Y. Chao, Q. He, B. Yang, H. Li, A. Borisevich, S. Dai, *Small* **2016**, *12*, 3535-3542.

17 (*a*) K. Yan, H.-W. Lee, T. Gao, G. Zheng, H. Yao, H. Wang, Z. Lu, Y. Zhou, Z. Liang, Z. Liu, S. Chu, Y. Cui, *Nano Lett.* **2014**, *14*, 6016-6022; (*b*) L. Shi, A. Xu, T. Zhao, *ACS Appl. Mater. Interfaces* **2017**, *9*, 1987-1994.

18 W. Luo, L. Zhou, K. Fu, Z. Yang, J. Wan, M. Manno, Y. Yao, H. Zhu, B. Yang, L. Hu, *Nano Lett.* **2015**, *15*, 6149-6154.

19 M.-P. Gelin, B. Ameduri, *J. Polym. Sci., Part A: Polym. Chem*. **2003**, 41, 160-171.

20 V. Aravindan, P. Vickraman, *J. Appl. Polym. Sci.* **2008**, *108*, 1314-1322.

21 (*a*) Y. Zhu, F. Wang, L. Liu, S. Xiao, Z. Chang, Y. Wu, *Energy Environ. Sci.* **2013**, *6*, 618-624; (*b*) C. S. Kim, S. M. Oh, *Electrochim. Acta* **2000**, *45*, 2101-2109.

22 J. Shim, J. S. Lee, J. H. Lee, H. J. Kim, J.-C. Lee, *ACS Appl. Mater. Interfaces* **2016**, *8*, 27740-27752.

23 N.-S. Choi, S.-W. Ryu, J.-K. Park, *Electrochim. Acta* **2008**, *53*, 6575-6579.

24 H. J. Kim, M.-Y. Lim, K. H. Jung, D.-G. Kim, J.-C. Lee, *J. Mater. Chem. A* **2015**, 3, 6798-6809.

25 X. Hao, J. Zhu, X. Jiang, H. Wu, J. Qiao, W. Sun, Z. Wang, K. Sun, *Nano Lett.* **2016**, *16*, 2981-2987.

26 X.-B. Cheng, T.-Z. Hou, R. Zhang, H.-J. Peng, C.-Z. Zhao, J.-Q. Huang, Q. Zhang, *Adv. Mater.* **2016**, *28*, 2888-2895.

27 G. Zheng, C. Wang, A. Pei, J. Lopez, F. Shi, Z. Chen, A. D. Sendek, H.-W. Lee, Z. Lu, H. Schneider, M. M. Safont-Sempere, S. Chu, Z. Bao, Y. Cui, *ACS Energy Lett.* **2016**, *1*, 1247-1255.

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Highly dendrite-suppressing gel polymer electrolytes for lithium metal batteries are presented utilizing perfluoropolyether-functionalized 2D boron nitride nanoflakes as a multifunctional additive.